Huawei Access Node QoS/Congestion Event Troubleshooting and Resolution Standard Operating Procedure

Node: YEL-ACC-67 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_2216

Description

The occurrence of QoS/Congestion events, such as microbursts, interface tail drops, and QoS queue starvation, on the Huawei Access Node (YEL-ACC-67) at the Yelahanka Sub-station, can be attributed to a complex interplay of various factors. This includes the inherent limitations of the Huawei QoS implementation, which utilizes a combination of Priority Queuing (PQ), Class-Based Queuing (CBQ), and Weighted Round Robin (WRR) scheduling algorithms to manage traffic prioritization and congestion control. Furthermore, the Access Node's role in aggregating traffic from multiple downstream devices and forwarding it to the upstream network can exacerbate congestion issues, particularly during periods of high network utilization or when traffic patterns exhibit bursty characteristics. The critical severity of the ALARM_2216 indicates a significant impact on network performance, necessitating prompt troubleshooting and resolution to mitigate the effects of congestion and ensure reliable network operation.

Pre-requisite Checks

- Dispatch remote hands to the Yelahanka Sub-station to perform on-site checks
- Verify redundant link status to ensure no single point of failure
- Confirm all necessary tools and equipment are available for troubleshooting and repair

Potential Root Causes

Cause	Probability
Insufficient QoS configuration or policy application	High
Interface congestion due to high network utilization or bursty traffic patterns	Medium
Hardware or software issues affecting the Access Node's forwarding performance	Low

Diagnostic Commands & Expected Outputs

```
Command: display qos policy, Expected Output: QoS policy configuration and application status, Description: This command isolates QoS policy configuration issues by displaying the current QoS policy settings and their application status on the Access Node. It helps to identify if QoS policies are correctly configured and applied to the relevant interfaces.

Command: display interface brief, Expected Output: Interface status, including traffic statistics and error counts, Description: This
```

```
command provides a concise overview of interface status, including traffic statistics, error counts, and other relevant metrics. It aids in identifying interfaces experiencing congestion or errors, which can contribute to QoS/Congestion events.

Command: display cpu-usage, Expected Output: CPU utilization statistics, Description: This command displays CPU utilization statistics, helping to identify if high CPU usage is contributing to forwarding performance issues or QoS/Congestion events on the Access Node.
```

Resolution Steps

Step 1: Step: Apply QoS Policy, Command: qos policy policy_name, Description: Apply a predefined QoS policy to the affected interface or globally on the Access Node to ensure proper traffic prioritization and congestion control. This involves specifying the policy name and configuring the necessary QoS parameters, such as queue scheduling algorithms and bandwidth guarantees.

Step 2: Step: Adjust Interface Settings, Command: interface gigabitethernet 0/0/1, Description: Enter the interface configuration mode for the affected interface and adjust settings as necessary to alleviate congestion. This may include tweaking the interface's MTU, adjusting the queue sizes, or applying traffic shaping and policing to manage bursty traffic patterns.

Step 3: Step: Upgrade Software, Command: boot-loader file flash:/huawei_vrpxxx.bin, Description: If hardware or software issues are suspected, upgrade the Access Node's software to the latest version to ensure that any known bugs or performance issues are addressed. This involves specifying the location and filename of the new software image and initiating the boot process with the upgraded image.

Post-Check Verification

```
Command: display qos statistics, Expected Output: QoS statistics, including queue depths and drop counts, Description: This command displays QoS statistics, allowing verification that the applied QoS policies are effective in managing congestion and that queue depths and drop counts are within acceptable thresholds.

Command: display interface traffic, Expected Output: Detailed traffic statistics for the interface, Description: This command provides detailed traffic statistics for the specified interface, enabling verification that traffic patterns are no longer contributing to congestion events and that the network is operating within expected parameters.
```

Roll-back Procedures

- Step: Revert QoS Policy, Command: undo qos policy policy_name, Description: If the applied QoS policy does not alleviate the congestion issue or introduces unexpected behavior, revert the QoS policy to its previous state to maintain network stability and prevent further disruptions.

- Step: Restore Interface Settings, Command: default interface gigabitethernet 0/0/1, Description: If adjustments to the interface settings do not resolve the issue or cause unforeseen problems, restore the interface configuration to its default state to ensure that any changes made during troubleshooting are reversed, thereby minimizing potential impact on the network.

- Step: Revert Software, Command: boot-loader file flash:/previous_huawei_vrpxxx.bin, Description: In the event that a software upgrade fails to resolve the issue or introduces new problems, revert the Access Node to its previous software version to maintain network stability and continuity. This involves specifying the location and filename of the previous software image and initiating the boot process with the downgraded image.